# RATE-DISTORTION LOSS ATTRIBUTION FOR AV1 INTRA PARTITION DECISIONS

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## ABSTRACT

The AOMedia Video 1 (AV1) codec splits each intra-frame super block by a recursive search over ten partition shapes per node, and learned pruners are the dominant strategy to shorten it. Such pruners couple the information that scores a node with the aggressiveness of the policy acting on it, so their time savings do not identify which information carries the decision. This paper presents an offline attribution protocol that fixes the pruning policy and reads the rate-distortion loss of competing representations at matched search-cost reduction. Over 3.81 million held-out decision nodes, twenty-four compact descriptors reduce the loss of isolated block variance by 7.0 times, eight causal partitioning columns by a further 1.60 times, and four quantization and position columns add nothing. A ConvNeXt of 28.1 million parameters reading raw pixels stays 4.1 times behind. The loss is ordinal and does not convert into Bjontegaard delta bitrate. To the best of the authors' knowledge, this is the first attribution of rate-distortion loss to information access in AV1 partitioning.

***Index Terms***— Video coding, AV1, intra-frame partitioning, feature representation, model capacity, machine learning

## 1. INTRODUCTION

Video coding removes the spatial and statistical redundancy of a raw sequence so that it can be stored and transmitted at a small fraction of its original rate. Every coding choice trades bitrate against reconstruction fidelity, and encoders resolve that trade-off by rate-distortion optimization, selecting at each decision the candidate of lowest Lagrangian cost [1]. Coding efficiency is the fidelity obtained at a given rate, and it degrades whenever a decision is taken without evaluating the candidate the full search would have selected. That degradation, and not encoding time, is what this work measures.

Two families of codecs pursue that efficiency. The standardization path produced High Efficiency Video Coding [2] and Versatile Video Coding [3], both from ISO/IEC MPEG and ITU-T VCEG, while the Alliance for Open Media released the AOMedia Video 1 (AV1) in 2018 as an open and royalty-free codec [4], drawing on VP9, Thor, and Daala.

AV1 follows the hybrid block-based coding flow, combining intra-frame and inter-frame prediction, transforms, quantization, in-loop filters, and entropy coding, introducing new tools at every one of those stages [5, 6]. Among them, the partition structure was extended to ten shapes per node, against the four of VP9. That coding efficiency is paid for in computational effort: comparative studies report AV1 encoding times far above those of its competitors [7, 8], the gap widening at the highest quality configuration.

Inside intra-frame coding, the recursive partition search is the single most expensive stage at that configuration. The search visits a quad-tree of square nodes, evaluates up to ten shapes at each, and descends into every node it does not resolve, so cost grows with the number of nodes rather than with the difficulty of the content.

Several works attack this cost with learned decisions. In AV1, tool selection guided by user priority and by coding efficiency was proposed in [9, 10], a fast intra mode decision based on the accuracy of a rate-distortion model in [11], and decision trees for inter prediction in [12]. The same problem in Versatile Video Coding was addressed with light gradient boosting [13], random forests [14], and variance and gradient heuristics [15], while dedicated hardware architectures targeted the AV1 intra prediction modes [16, 17]. These works can be classified as coding-efficiency oriented [10, 11, 13–15], hardware oriented [16, 17], or aimed at a different coding stage [9, 12]. All report time savings under a policy of their own choosing, and none separates the quality of the scoring source from the aggressiveness of the policy, nor measures which information is decisive for the partition choice.

This paper presents an offline attribution protocol for AV1 intra partitioning. The main strategy of this work is to hold the pruning policy and the amount of skipped search work fixed, and to vary only what a representation is allowed to see, so that the difference in rate-distortion loss is attributable to information access alone. Under this protocol, compact descriptors outperform a high-capacity convolutional model reading raw pixels, and the causal partitioning neighborhood carries decision information that luminance does not provide.

## 2. INTRA PARTITION SEARCH IN AV1

AV1 partitions each frame into super blocks of $128\times128$ samples, which the encoder may alternatively set to $64\times64$ [5]. Each super block is recursively divided by a search over a quad-tree of square nodes that may descend to $4\times4$ samples. At every node the encoder may evaluate ten partition shapes, organized in four groups: the undivided block `PARTITION_NONE`, the quaternary `PARTITION_SPLIT`, the two rectangular shapes, and the six extended shapes, four AB and two four-way.

The evaluation order inside a node is fixed: `PARTITION_NONE` first, then `PARTITION_SPLIT`, whose opening triggers the recursion into the four children, then the rectangular shapes, and the extended ones last.

This order defines when each kind of information becomes available, the property this work exploits. Before the search of a node begins, the encoder already holds the source luminance of the block, the frame quantization index, the node position, and the partition decisions of the causal neighbors above and to the left, but no rate-distortion cost for the node yet. Fig. 1 summarizes this timeline.

The action studied in this paper is the earliest one available:

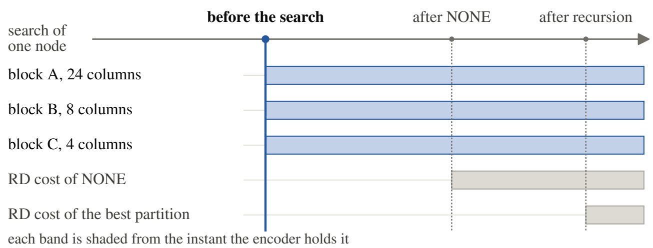


**Fig. 1**. Information availability inside the search of one node: search time on the horizontal axis, information item on the vertical.

committing the node to the undivided shape before any rate-distortion evaluation is paid, skipping every remaining candidate of the node and the subtree below it. Section 3 measures its two consequences: the search work no longer performed, and the rate-distortion cost lost when the discarded subtree was optimal.

This work is focused on that commitment at the pre-search insertion point of the intra-frame partition tree. Inter-frame prediction, transform type selection, quantization, in-loop filters, and entropy coding are not included in the loop attacked here, and therefore they are not focused on this work. The decision at the $128\times128$ root is also outside this scope: the protocol attaches at the four $64\times64$ quadrants of each super block and models the 64, 32, and 16 sample levels. Blocks of $8\times8$ samples were never split in this dataset, no divided label among 16.91 million, so $8\times8$ is the effective leaf; they are excluded from modeling but kept in the cost accounting.

## 3. PROPOSED ATTRIBUTION PROTOCOL

The proposed protocol compares representations by what they are allowed to see, under one fixed pruning policy and at a matched amount of skipped search work.

### 3.1. Representations Declared by Information Access

The representations are declared by information access rather than by architecture. The first is the isolated variance of the block, the classical descriptor of partition heuristics, scored by a non-learned decreasing map of the variance under one threshold shared by the three levels, so the sweep of $\tau$ traverses every commitment set a variance threshold can produce. The second, denoted A, is a twenty-four column vector of compact luminance descriptors: second-order and gradient statistics of the block and of its quadrants, edge measures, and the contrast against its parent and sibling blocks. Three of those columns do not derive from luminance, namely the normalized quantization index and the two coordinates of the node inside the 64-sample unit.

Block B adds eight columns of causal partitioning neighborhood: availability of the above and left neighbors, their already decided widths and heights in logarithmic scale, and the relative granularity and anisotropy of the neighborhood. Block C adds four columns of effective quantization and position, namely the dequantization step of the direct-current coefficient, the normalized position within the frame, and the node depth. Four tabular representations are built from these blocks: A, A+B, A+C, and A+B+C, the last holding thirty-six columns. All of these columns are natively available during encoding at the pre-search instant, ensuring zero computational overhead for feature extraction.

The remaining representation is the deep pixel arm, a multi-level convolutional model of the ConvNeXt family [18] mirroring the partition hierarchy. Its input is a two-channel tensor formed by the luminance of the $64\times64$ unit and a constant quantization plane. Three stages of its trunk are merged top-down, so each cell of the resulting map holds local detail and the context of the whole unit. This arm tests whether raw pixels plus capacity suffice; it is not an architectural baseline. A random scorer is included as a control.

Each of the four tabular representations is read by three multilayer perceptrons, one per block size, with two hidden layers of 64 and 32 rectified units, trained with hard-label cross entropy and AdamW [19], learning rate $1\times10^{-3}$, batches of 4096, and a fixed budget of 30 epochs. All four share this training strategy and differ only in which columns they read, so the verdict falls on information rather than on capacity or on how the models were trained. Each is trained with three seeds and reported as their mean.

### 3.2. Policy, Cost Model, and Loss

The policy is fixed for every representation. A node $n$ is committed to `PARTITION_NONE` whenever the probability the model assigns to that class exceeds a confidence threshold $\tau$, in which case the node is charged only its own `PARTITION_NONE` evaluation and the recursion stops; otherwise the node is evaluated in full and the search descends. The rate-distortion loss charged to a committed node is the overhead of that commitment against the partition the full search selected,

$$r(n) = J_{\mathrm{none}}(n) - J^*(n), \tag{1}$$

where $J_{\mathrm{none}}(n)$ is the rate-distortion cost of `PARTITION_NONE` and $J^*(n)$ is the cost of the partition the reference search chose for $n$. It is non-negative by construction, and zero whenever the reference decision was the undivided shape. No encoding is repeated: the protocol is a counterfactual replay over the log of one exhaustive search.

Search work is counted analytically rather than timed. Each node is charged the number of shape candidates a full search would evaluate there, weighted by the area of the block: nine candidates for blocks of 64, 32, and 16 samples, and four for blocks of 8 samples, which admit neither AB nor four-way shapes. `PARTITION_SPLIT` is not among the nine: its cost is the recursion into the four children, already charged there. The cost reduction $\Delta(\tau)$ is the percentage difference between the work performed by the policy and by the full search. It is a counted quantity, not wall-clock time.

Summing (1) over the committed nodes gives the counterfactual loss of the pruning under that replay,

$$L(\tau) = 100\,\frac{\sum_{n\in P(\tau)} r(n)}{\sum_{n\in N} J_{\mathrm{none}}(n)}, \tag{2}$$

where $P(\tau)$ is the set of nodes committed at threshold $\tau$ and $N$ is the set of all decision nodes of the four quantization points. The denominator is accumulated before any threshold is fixed and stays constant along the sweep, so $L$ tends to zero with the pruning.

Three properties of $L$ restrict its reading, stated before any number is reported. First, the denominator aggregates the undivided costs of the three hierarchy levels, which cover the same image area, so it exceeds the cost of coding that area. This deflates the ratio for a given numerator, and nothing more: it licenses no comparison between the magnitude of $L$ and any loss measured by re-encoding. Second, the replay discounts recorded values, whereas pruning inside the encoder also alters the reconstruction that serves as reference to later blocks, a propagation absent here. Third, the sum runs over four quantization points, whose Lagrangian multipliers differ, so the aggregate is weighted toward the coarsest of them, where rate-distortion costs are largest. Consequently, no conversion factor between $L$ and the Bjøntegaard delta bitrate [20] is postulated, and $L$ is used strictly in an ordinal regime, every ratio being taken between representations read at the same operating point.

Lowering $\tau$ commits more nodes and raises $\Delta$. It need not raise $L$, since committing a node ends the recursion and the committed sets are therefore not nested, though every curve measured here is monotone. Each representation is a curve rather than a point, compared by reading the curves at the same cost reduction. The reference reading here is $\Delta = 25\%$, one quarter of the search work skipped.

### 3.3. Dataset and Experimental Setup

The reference labels were extracted from libaom v3.10.0 [21], instrumented under a compile-time guard so the production binary remains byte-identical to the original. The encoder was configured for All-Intra following the AOM Common Test Conditions [22] at `cpu-used=0`, which enables the full rate-distortion search and makes the recorded decisions optimal by construction.

The experiments use sixteen sequences of the UVG dataset [23] at 4K resolution (3840×2160), 8 bits and 4:2:0 chroma subsampling. The restriction to 4K is deliberate. Search cost grows with the number of nodes, so this is the resolution at which the exhaustive search is most prohibitive, and resolution also governs the super block size, so averaging over resolutions would average over tree depths. Each sequence was encoded at four quantization points (`cq-level` 20, 32, 43, and 55) over five frames sampled uniformly along the clip, since consecutive frames are nearly identical under All-Intra. The resulting dataset holds 26.98 million samples, of which 10.07 million are decision nodes. The split is by sequence and was frozen before any measurement: ten sequences for training, three for validation, three for the reserved test set. No video sequence was reused across the training, testing, or evaluation datasets, thus avoiding data leakage and overfitting. All results reported here are measured on the six held-out sequences, which contribute 226 447 units of 64 samples and 3 808 703 decision nodes. The models were implemented in PyTorch [24].

### 3.4. Fair Treatment of the Deep Arm

A negative result on the deep arm only counts if it was given every advantage. Three controls were applied.

The first is exposure. The arm was trained on the same training set as the tabular representations, and its checkpoint was selected by validation loss over three of the six held-out sequences on which every arm is later read, whereas the tabular arms run a fixed epoch budget and select nothing. The asymmetry favors the deep arm, which these results rule against. The second is capacity. The fusion width was doubled from 128 to 256 channels, and the wider model is 1.0% worse in validation loss.

The third is the training objective. A plain cross entropy (CE) and a cost-sensitive variant [25], which weights each node by the rate-distortion loss its misclassification would incur, were trained on the same data, schedule, and selection criterion. The cost-sensitive objective is better at every operating point from 10% of cost reduction onward and is the one reported in Section 4, so the deep arm is read at its better objective.

**Table 1**. Rate-Distortion Loss at Matched Cost Reduction, in Parts per Million of the Aggregated Undivided Cost

| **Representation** | **10%** | **15%** | **20%** | **25%** | **30%** |
|---|---|---|---|---|---|
| Random control | 1963 | 2989 | 4066 | 5166 | 6342 |
| Variance | 43 | 127 | 250 | 374 | 555 |
| ConvNeXt, plain CE | 65 | 105 | 164 | 272 | 442 |
| ConvNeXt, width 256 | 75 | 122 | 194 | 294 | 444 |
| ConvNeXt, cost-sensitive | 54 | 80 | 140 | 219 | 379 |
| A (24 columns) | 5 | 13 | 28 | 53 | 91 |
| A+C (28 columns) | 5 | 12 | 27 | 64 | 122 |
| A+B+C (36 columns) | 2 | 6 | 18 | 40 | 78 |
| A+B (32 columns) | **2** | **5** | **15** | **33** | **69** |

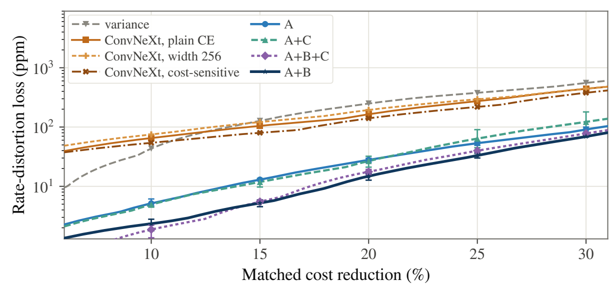


**Fig. 2**. Rate-distortion loss against matched cost reduction, logarithmic vertical axis. Bars give the amplitude over three seeds. The random control of Table 1 is omitted, as it alone spans a decade of the axis.

## 4. RESULTS AND DISCUSSION

Table 1 reports the loss $L$ of (2) over the six held-out sequences at five matched cost reductions, in parts per million, lower being better, and Fig. 2 shows the corresponding frontiers.

The main conclusion when observing Table 1 is that the four tabular representations stay below every pixel-domain arm at every cost reduction reported, so the separation between the families is not an artifact of one operating point. At 25%, isolated variance loses 374 ppm while the twenty-four compact columns of A lose 53 ppm, a reduction of 7.0 times, against the 5166 ppm of the random control. This contradicts any reading in which the pixel domain would be exhausted by variance. Unlike the ratios among the tabular representations, this 7.0-times margin also absorbs the per-level fitting a single variance threshold cannot express, and is therefore a margin over the deployed heuristic rather than an attribution to the extra columns alone.

Adding the eight causal partitioning columns to A lowers the loss from 53 to 33 ppm, a further reduction of 1.60 times. The separation is robust to initialization: the worst seed of A+B, at 36 ppm, still beats the best seed of A, at 49 ppm. This is the central result: those eight columns describe the partition choices of the already coded neighbors, are not computable from the block luminance under any transformation, and therefore carry information produced by the encoding process itself.

Conversely, the four columns of effective quantization, frame position, and node depth add nothing. A+C reaches 64 ppm against 53 for A at 25%, with a seed amplitude of 45 ppm against 8 for A, so the two distributions overlap: block C destabilizes the fit without adding information. Combined with the causal columns the effect is unambiguous: A+B reaches 33 ppm against 40 for A+B+C, and all nine pairwise seed comparisons agree. Fig. 2 places the crossover at 14.2%: below it the order reverses by under 1.1 ppm, above it A+B leads by a widening margin. The thirty-six column vector is therefore not the offline optimum in the regime read here, though A+B's superiority has not been verified inside the encoder.

The deep arm does not support the hypothesis that raw pixels plus capacity suffice. In its most favorable configuration it reaches 219 ppm at 25%, 4.1 times worse than A, although it holds 28 128 638 parameters against the 11 337 of the three networks of A, a factor of 2481. Doubling the fusion width degrades the result rather than improving it, at 294 ppm against 272 under the same objective, and below 11% of cost reduction the whole deep family is beaten by isolated variance. This may be attributed to missing information rather than to available capacity, the arm reading the same pixels as A; it is therefore no upper bound of the pixel domain, since a model outperformed by one with strictly smaller access establishes no ceiling.

Three limitations qualify these results. The three seeds give an amplitude, not a confidence interval. The objective ablation covers two objectives, and the strength of the cost weighting was not swept. And the dataset is 4K only, so whether the causal neighborhood weighs the same where the super block is 64×64 is not established.

## 5. CONCLUSIONS

This paper presented an offline attribution protocol that measures the rate-distortion loss of competing representations of the AV1 intra partition decision at matched search-cost reduction, concluding that the encoder's causal state carries decision information luminance alone does not provide. It holds the pruning policy and the training strategy fixed, varying only the information a representation may read, and was evaluated over 3.81 million held-out decision nodes of six UVG 4K sequences encoded with libaom v3.10.0. Twenty-four compact descriptors improve on isolated variance by 7.0 times, eight causal partitioning columns on those by a further 1.60 times, and four quantization and position columns add nothing. A ConvNeXt with 28.1 million parameters remains 4.1 times behind, and doubling its width makes it worse, so capacity is not the binding constraint. Besides, to the best of the authors' knowledge, this is the first attribution of rate-distortion loss to information access in AV1 partitioning.

## 6. ACKNOWLEDGMENT

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